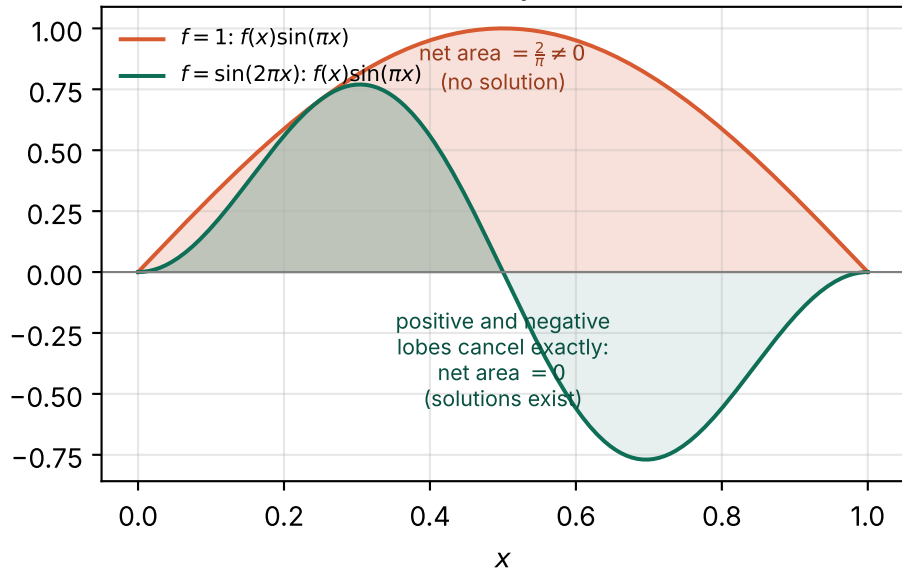


solvability test: $\int_0^1 f(x)\sin(\pi x) dx$



every $u_c = \frac{\sin(2\pi x)}{3\pi^2} + c\sin(\pi x)$ solves it

